

ENTRY 060 Sherbrooke Chain-Topology Generation, Fixed — September 1, 2026 — Status: COMPLETE

Entry 060 — Diagnosing and Fixing the Entry 058 Stall

Sherbrooke Chain Circuits Went From Zero to Twelve, and the Fix Also Helped Kyiv

The Problem

Entry 058 introduced chain-topology GHZ circuits ($q_0 \text{---} q_1 \text{---} \dots \text{---} q_k \text{---} 1$, entangled sequentially) as a second large-circuit shape alongside the existing hub-and-spoke star topology. Generation worked for Kyiv but stalled completely on Sherbrooke — two consecutive full 150-second time budgets produced zero Sherbrooke chain circuits, and the entry shipped with chain data Kyiv-only, an honestly logged gap.

Root Cause

Entry 058's generator sampled k qubits uniformly at random from the whole chip and chained them in that random order. Most consecutive pairs in the resulting chain are therefore physically far apart on the real coupling map, so after SWAP routing the circuit's entangling structure is not spatially local. The Aer matrix-product-state simulator is fast specifically because it exploits 1D-local entanglement — feed it a routed circuit that isn't spatially local and its bond dimension blows up. Isolated timing tests confirmed this directly: a single $k=10$ Sherbrooke trial with a random-order chain measured 48–120+ seconds, well past the 150-second total budget per chip. Kyiv's random samples happened to land local more often by chance in Entry 058's seed, which is why Kyiv appeared to work at all — the underlying bug was present there too.

The Fix

Build each chain by walking the real coupling graph itself: pick a random start qubit, then repeatedly step to a random unvisited physical neighbor, falling back to the nearest unvisited qubit by BFS distance when the walk boxes itself in. This is still a randomized chain — the start point and walk choices are seeded and vary run to run — but it stays spatially local on the chip, restoring the property that made MPS simulation fast in the first place. Measured effect on a single Sherbrooke $k=10$ trial: 48s → 0.5s, roughly a 90x speedup.

Results

With the fix, both chips generated their full target (4 circuits each at $k=10/12/14$) in under 6 seconds total per chip, versus the previous two full-budget timeouts producing zero on Sherbrooke. The chain-topology dataset grew from 4 circuits (Kyiv-only) to 24 (12 per chip), and the combined graph dataset grew from 3,221 to 3,241 circuits. Capacity headroom was unaffected — max nodes/edges across the new data (67/106) stayed well within the existing 90/130 padding limits, so no retraining-capacity bump was needed.

	Entry 058 (Kyiv-only chain)	Entry 060 (both chips)
	3,221	3,241
	4 (Kyiv only)	24 (12 Kyiv, 12 Sherbrooke)
	1.09 / 0.977	1.16 / 0.970
lapse MAE	0.63	2.15
MAE / R ² / fc_MAE	3.55 / 0.859 / 7.02	3.69 / 0.839 / 7.40
MAE / R ² / fc_MAE	2.88 / 0.901 / 4.69	2.98 / 0.879 / 2.45

Reading the Results Honestly

This is not a uniform win, and it is reported as such. Sherbrooke→Kyiv floor-collapse MAE improved substantially (4.69 → 2.45 points) — training on Sherbrooke's new chain data measurably helped the model handle floor-collapse cases when tested cold on Kyiv. But same-chip MAE, same-chip floor-collapse MAE, and Kyiv→Sherbrooke MAE/R² all moved slightly in the worse direction. The most likely explanation is not model

regression but a harder, more honest test set: the Kyiv→Sherbrooke and same-chip evaluations now include the 12 new Sherbrooke chain circuits for the first time, several of which are genuine floor-collapse cases (aer_ground_truth as low as 6–14%) that simply did not exist in the Sherbrooke test pool before. A test set that got harder because it got more representative is not the same thing as a model that got worse, but the two are easy to conflate, so both readings are recorded here rather than only the favorable one.

Entry 060 — Conclusion

The Sherbrooke chain-generation gap flagged as unresolved in Entry 058 is closed: both chips now have symmetric star and chain large-circuit coverage, diagnosed via direct timing measurement rather than guesswork, and fixed with a 90x generation speedup that also quietly benefited Kyiv. Cross-chip results are mixed rather than uniformly better, and are reported that way — the clearest gain is Sherbrooke→Kyiv floor-collapse MAE, while Kyiv→Sherbrooke got a harder, more representative test set rather than a worse model.

Research Log — Index Addendum (Entry 060)

Files: `entry060_chain_circuits.py` (fixed generator), `entry060_build_graphs.py`, `entry060_train_gnn.py`, `entry060_cross_chip.py`. Data: `entry058_chain_dataset.json` (24 records, same file, both entries append), `entry060_graph_dataset.json` (3,241 graphs), `entry060_gnn_results.json`, `entry060_cross_chip_results.json`.